

# Chapter 1: Cellular Respiration & ATP Synthesis

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## 1.1 Overview of Cellular Energy Harvest

### Catabolic Oxidation of Glucose to ATP

Cellular respiration is the catabolic biochemical pathway by which living cells harvest chemical energy stored in glucose molecules to generate adenosine triphosphate (ATP), the universal energy currency of biological work.

The overall stoichiometric equation for aerobic cellular respiration is:  $C_6H_{12}O_6 + 6 O_2 \rightarrow 6 CO_2 + 6 H_2O + \sim 30\text{-}32 \text{ ATP} + \text{Heat}$

## 1.2 The Four Sequential Stages

### Glycolysis, Pyruvate Oxidation, Krebs Cycle, and Oxidative Phosphorylation

1. Glycolysis: Occurs in the cytoplasm. Splits 1 glucose (6C) into 2 pyruvates (3C), yielding a net gain of 2 ATP and 2 NADH without requiring oxygen. 2. Pyruvate Oxidation: Pyruvate translocates into the mitochondrial matrix, converting to Acetyl-CoA and releasing  $CO_2$  and NADH. 3. Citric Acid (Krebs) Cycle: Acetyl-CoA combines with oxaloacetate (4C) to form citrate (6C). Through cyclical oxidation steps, it generates 6 NADH, 2  $FADH_2$ , 2 ATP, and 4  $CO_2$  per glucose. 4. Oxidative Phosphorylation: High-energy electrons donated by NADH and  $FADH_2$  cascade through complexes I-IV in the inner mitochondrial membrane, driving protons ( $H^+$ ) into the intermembrane space.

## 1.3 Chemiosmosis & ATP Synthase

### The Proton Motive Force

Peter Mitchell's Chemiosmotic Hypothesis explains that the electrochemical proton gradient across the inner mitochondrial membrane (the proton motive force) drives the mechanical rotary motor of ATP Synthase, condensing  $ADP + P_i$  into ATP.

### Key Takeaways:

- Glycolysis occurs anaerobically in the cytosol, yielding net 2 ATP and 2 NADH.
- Krebs cycle completes the total oxidation of organic carbons to  $CO_2$  in the mitochondrial matrix.
- Oxygen serves as the terminal electron acceptor in the electron transport chain, forming  $H_2O$ .
- Chemiosmosis harnesses the proton gradient to drive ATP synthase.